P-A-86a

AUGUST 14, 1957

SUPERSEDING
Int. Fed. Spec. P-A-0086 (GSA-FSS)
April 24, 1956, and
Fed. Spec. P-P-156
October 18, 1945

FEDERAL SPECIFICATION

ABRASIVE PASTE, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers dental abrasive pastes, suitable as abrading and polishing agents for plastic and porcelain teeth, and for cast metal restorations.

1.2 Classification.—Dental abrasive pastes covered by this specification shall be of the following types, as specified (see 6.2):

Type I.—Silicon-carbide paste.

Type II.—Aluminum-oxide paste.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

RR-S-366—Sieves, Standard for Testing Purposes.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585—Boxes; Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-676 — Boxes, Set-Up, Paperboard.

PPP-M-186 — Medicinal Products and Clinical Laboratory Reagents: General Specification for Containers (Packaging and Packing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Composition.

3.1.1 *Type I, silicon-carbide paste.*—Type I paste shall consist essentially of silicon carbide incorporated in glycerine, and shall contain not less than 60 percent by weight of silicon carbide. Gums or colloidal matter may be added to the vehicle, as necessary, to obtain the desired consistency. The paste shall contain no harmful ingredients, shall be neutral or alkaline in reaction, and be pleasant to the taste.

3.1.2 *Type II, Aluminum-oxide paste.* — Type II paste shall consist essentially of aluminum oxide formulated in a nonwater soluble base, such as petrolatum. The paste shall contain no harmful ingredients, and be pleasant to the taste.

3.2 Description.

3.2.1 *Silicon-carbide paste.* — Silicon-carbide paste shall be suitable for intra- or extra-oral grinding of porcelain teeth. Paste shall be capable of being washed off the teeth with warm water.

3.2.2 *Aluminum-oxide paste.*—Aluminum-oxide paste shall be an all purpose paste for intra- or extra-oral milling, finishing, and polishing of plastic teeth, and for cast metal restorations. Paste shall be of such consistency that it can be removed from restorations by brushing with soap and warm water. When used for intra-oral purposes, it shall be possible to remove the paste with warm water or mouth wash.

3.3 Preservative.—Abrasive pastes shall contain a preservative which is nontoxic and pleasant to the taste.

3.4 Greases.—Silicon-carbide paste shall be free from greases or oils.

3.5 Fineness.

3.5.1 *Silicon carbide.*—One hundred percent of silicon carbide shall pass through a No. 100 sieve, and not less than 80 percent shall pass through a No. 200 sieve. Sieves shall conform to Federal Specification RR–S–366.

3.5.2 *Aluminum oxide.* — Paste shall be formulated of 150-mesh aluminum oxide.

3.6 Consistency.—Abrasive pastes shall be homogeneous in consistency. When extruded from the tube on to a glass plate, the paste shall retain its shape. Consistency shall also be such as to permit extrusion from tube without too much pressure, and to maintain original uniform dispersion of abrasive, and prevent slow settling out or concentration of abrasive in one portion of tube on storage.

3.7 Lot or control number.—Each tube of abrasive paste shall be marked with a lot or control number.

3.8 Instructions.—Adequate instructions for specific uses and technique for use, of each abrasive paste, shall be furnished with each unit.

3.9 Workmanship.—Workmanship shall be first class throughout. The ingredients shall be so incorporated as to insure a uniformly, smooth, homogeneous finished product.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance

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with instructions furnished by the procuring activity (see 6.2). In addition, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL-STD-105.

4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Tests.**—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*—Three ounces of abrasive paste, of the stock number specified, constitute one unit.

TABLE I.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Shipping container
6520-500-0300 or 6520-500-0400	1 unit	6 units	96 units

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Immediate container.*—Each unit shall be packaged in a collapsible metal tube of suitable size and design. Tube shall be fitted with a suitable metal or plastic screwcap, in accordance with Federal Specification PPP-M-186.

5.1.3.2 *Individual carton.*—Each tube shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-566 or PPP-B-676. Carton shall be adequately secured to prevent accidental opening.

5.1.3.3 *Intermediate package.*—The number of units specified in table I, shall be packaged in a carton of appropriate size and design, or shall be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use;

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PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I, or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* —The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit packages.*—Each tube shall be commercially identified. Each individual carton shall be marked with stock number, item identification, unit, quantity, lot (control) number, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data.—Purchasers should exercise any desired options and should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2).

(c) Sampling required (see 4.1).

(d) Inspection required (see 4.2).

(e) Place of final inspection (see 4.2).

(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

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6.5 **Transportation description.**—Transportation description applicable to this item is:

Buffing of polishing compounds, not otherwise indexed by name.

Carload minimum weight 36,000 pounds.
Motor volume minimum weight 36,000 pounds.

6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–500–0300 ABRASIVE PASTE, DENTAL, Silicon Carbide, 3 oz.

6520–500–0400 ABRASIVE PASTE, DENTAL, Alundum, 3 oz.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force—AF.

Copies of this specification may be purchased for .05 cents each.

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